Conversations for a Smarter Planet

How we can make our smart systems smarter.

The planet we live on is getting *smarter*. A world where digital intelligence can be embedded across entire systems, impacting everything from traffic flows to electric power to the way our food is grown, processed and delivered.

But you might be surprised to learn that information technology is itself in need of an intelligence makeover. It's not a problem with the technology per se. The problem is how all this technology is currently configured into systems. The fact is, the IT systems that underpin so much of how the world works must become much smarter.

How much smarter? The average commodity server rarely uses more than 6% of its available capacity, while in some organizations as many as 30% of servers aren't utilized at all. IT energy consumption is expected to double in the next five years. In some cases, nearly 70% of companies' IT budgets can be devoted to managing, maintaining, securing and upgrading their systems rather than building new capabilities, services and applications.

And consider what's coming: hundreds of billions of smart things – sensors, cameras, cars, shipping containers, intelligent appliances, RFID tags by the hundreds of millions – all becoming interconnected. This will enable new, highly flexible ways of interacting with customers, employees, patients and citizens from any device, anywhere. The resulting volume of data promises insight and intelligence to solve some of our biggest problems – but only if we can process and make sense of it in real time.

Fortunately, smarter computing models are at hand. With "service oriented" software, companies can unlock business services from the underlying technology, so their software can be changed and reused flexibly – at a fraction of the cost of developing it from scratch. Virtualization can help companies reinvent their data centers, eliminating up to 70% of their servers and 80% of their floor space. Service management software can orchestrate all of these systems from one place, while letting IT users serve themselves, cutting administrative costs. Together, these new capabilities enable "cloud computing", a new way of looking at IT as a distributed capability, which can be tapped into simply and easily.

Information technology has taken us a long way over the past 50 years. But seizing the opportunities before us will depend on more than intelligent machines. It will depend on spreading intelligence across our technology infrastructures. Let's build a smarter planet. Join us and see what others are thinking at **ibm.com**/change/in

To learn more about IBM's vision of a dynamic infrastructure, be a part of our DI Virtual Forum, scheduled for June 24th. You can register now at **ibm.com/systems/in/di**